

DBMS – SELLER AND INVENTORY MANAGEMENT

HUFT – E-Commerce Order Management Database System

1.DATABASE DESIGN

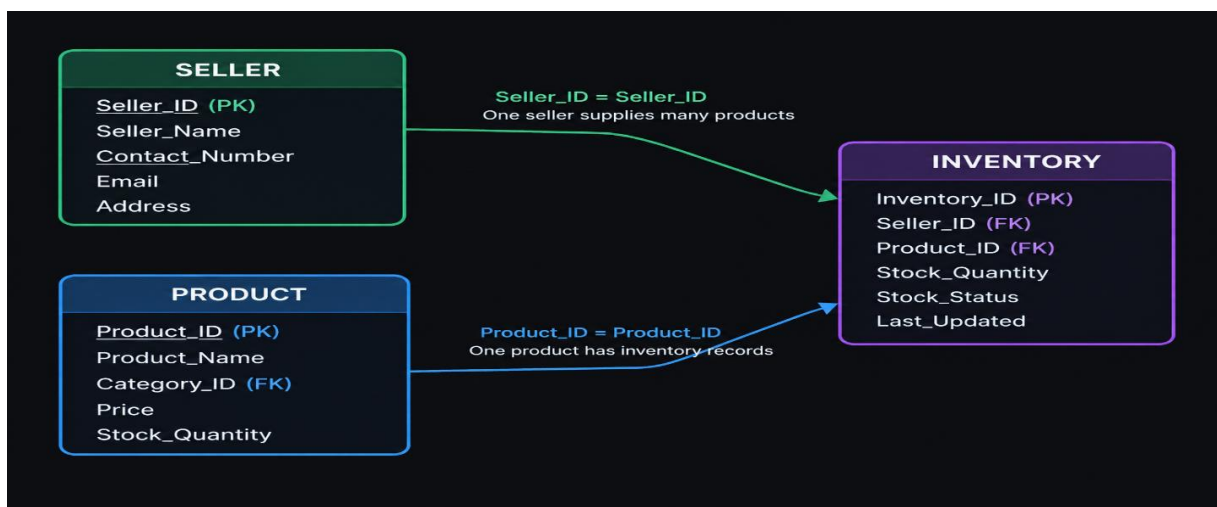
1.1 SELLER TABLE :

ATTRIBUTE	DESCRIPTION
Seller_ID	Unique seller ID
Seller_Name	Seller name
Contact_Number	Seller contact number
Email	Seller email address
Address	Seller address

1.2 INVENTORY TABLE :

ATTRIBUTE	DESCRIPTION
Inventory_ID	Unique inventory ID
Product_ID	Product ID
Seller_ID	Seller ID
Stock_Quantity	Available stock quantity
Stock_Status	Availability status of the product
Last_Updated	Date and time when inventory was last updated

2.TABLE RELATIONSHIP :



3. Table Creation

3.1 SELLER TABLE

```
SQL> CREATE TABLE HuftSeller (  
 2     Seller_ID NUMBER PRIMARY KEY,  
 3     Seller_Name VARCHAR2(100),  
 4     Contact_Number VARCHAR2(15),  
 5     Email VARCHAR2(100)  
 6 );
```

Table created.

```
SQL> ALTER TABLE HuftSeller  
 2 ADD Address VARCHAR2(200);
```

Table altered.

3.2 INVENTORY TABLE

```
SQL> CREATE TABLE HuftInventory (  
 2     Inventory_ID NUMBER PRIMARY KEY,  
 3     Product_ID NUMBER REFERENCES HuftProduct(Product_ID),  
 4     Seller_ID NUMBER REFERENCES HuftSeller(Seller_ID),  
 5     Stock_Quantity NUMBER NOT NULL,  
 6     Stock_Status VARCHAR2(20) NOT NULL,  
 7     Last_Updated DATE NOT NULL  
 8 );
```

Table created.

4. SELLER DATA INSERTION

```
SQL> INSERT INTO HuftSeller VALUES (1, 'Arun Pet Supplies', '9876543210', 'arunpets@gmail.com', 'Chennai');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (2, 'Pawsome Traders', '9876543211', 'pawsome@gmail.com', 'Tambaram');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (3, 'Happy Tails Store', '9876543212', 'happytails@gmail.com', 'Velachery');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (4, 'Pet Paradise', '9876543213', 'petparadise@gmail.com', 'Adyar');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (5, 'Furry Friends', '9876543214', 'furryfriends@gmail.com', 'Chrompet');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (6, 'Pet Care Hub', '9876543215', 'petcarehub@gmail.com', 'Guindy');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (7, 'Animal World', '9876543216', 'animalworld@gmail.com', 'Porur');
```

1 row created.

```
SQL> INSERT INTO HuftSeller VALUES (8, 'Pet Basket', '9876543217', 'petbasket@gmail.com', 'Anna Nagar');
```

1 row created.

5. INVENTORY DATA INSERTION

```
SQL> INSERT INTO HuftInventory VALUES (1, 101, 1, 50, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (2, 102, 2, 35, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (3, 103, 3, 25, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (4, 104, 4, 40, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (5, 106, 5, 30, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (6, 107, 6, 45, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.

SQL> INSERT INTO HuftInventory VALUES (7, 108, 7, 20, 'Available', TO_DATE('17-09-2026','DD-MM-YYYY'));
1 row created.
```

6. Seller product information

6.1 Seller and product details

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     p.Price,
  7     i.Stock_Quantity,
  8     i.Stock_Status
  9 FROM HuftSeller s
 10 JOIN HuftInventory i
 11 ON s.Seller_ID = i.Seller_ID
 12 JOIN HuftProduct p
 13 ON i.Product_ID = p.Product_ID
 14 ORDER BY s.Seller_ID;
```

SELLER_ID	SELLER_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK_QUANTITY	STOCK_STATUS
1	Arun Pet Supplies	101				

7.Inventory Status

7.1 Display Available products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM HuftInventory i
  8 JOIN HuftProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN HuftSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Available';
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
|      | 101
Chicken Dog Food
Arun Pet Supplies
|      | 50 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
|      | 102
Grain Free Cat Food
Pawsome Traders
|      | 35 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
|      | 103
Leather Dog Collar
Happy Tails Store
|      | 25 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
|      | 104
Pet Grooming Shampoo
Pet Paradise
|      | 40 Available
```

7.2 Display unavailable products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM HuftInventory i
  8 JOIN HuftProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN HuftSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Unavailable';
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
|  | 110
Pet Conditioner
Paw Mart
|  | 0 Unavailable
```

8.Inventory Status report

8.1 Complete inventory report

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status,
  7     i.Last_Updated
  8 FROM HuftInventory i
  9 JOIN HuftSeller s
 10 ON i.Seller_ID = s.Seller_ID
 11 JOIN HuftProduct p
 12 ON i.Product_ID = p.Product_ID
 13 ORDER BY i.Inventory_ID;
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  | 1
Arun Pet Supplies
Chicken Dog Food
|  | 50 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 2
Pawsome Traders
Grain Free Cat Food
|  |  | 35 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 3
Happy Tails Store
Leather Dog Collar
|  |  | 25 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 4
Pet Paradise
Pet Grooming Shampoo
|  |  | 40 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 5
Furry Friends
Cat Litter
|  |  | 30 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 9
Paw Mart
Pet Conditioner
|  |  | 0 Unavailable      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
|  |  | 10
Pet Needs
Cat Grooming Brush
|  |  | 28 Available      17-SEP-26
```

```
10 rows selected. |
```

8.2 Available and unavailable product count

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Total_Products
  4 FROM HuftInventory
  5 GROUP BY Stock_Status
  6 ORDER BY Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCTS
Available	9
Unavailable	1

8.3 Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM HuftInventory i
  8 JOIN HuftProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN HuftSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Available'
 13 ORDER BY p.Product_ID;
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
101	Chicken Dog Food	Arun Pet Supplies	50	Available

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
102	Grain Free Cat Food	Pawsome Traders	35	Available

8.4 Products with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM HuftProduct p
  8 JOIN HuftInventory i
  9 ON p.Product_ID = i.Product_ID
 10 JOIN HuftSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Quantity = 0;
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
110	Pet Conditioner	Paw Mart	0	Unavailable

8.5 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Stock_Quantity) AS Total_Stock
  5 FROM HuftSeller s
  6 JOIN HuftInventory i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY s.Seller_ID, s.Seller_Name
  9 ORDER BY s.Seller_ID;
```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
1	Arun Pet Supplies	50
2	Pawsome Traders	35

SELLER_ID	SELLER_NAME	TOTAL_STOCK
3	Happy Tails Store	25
4	Pet Paradise	

9. Inventory Summary

```
SQL> SELECT
  2     COUNT(*) AS Total_Inventory_Records,
  3     SUM(Stock_Quantity) AS Total_Stock,
  4     SUM(CASE
  5         WHEN Stock_Status = 'Available' THEN 1
  6         ELSE 0
  7     END) AS Available_Products,
  8     SUM(CASE
  9         WHEN Stock_Status = 'Unavailable' THEN 1
 10        ELSE 0
 11    END) AS Unavailable_Products
 12 FROM HuftInventory;
```

TOTAL_INVENTORY_RECORDS	TOTAL_STOCK	AVAILABLE_PRODUCTS	UNAVAILABLE_PRODUCTS
10	328	9	1